Tangents to a circle

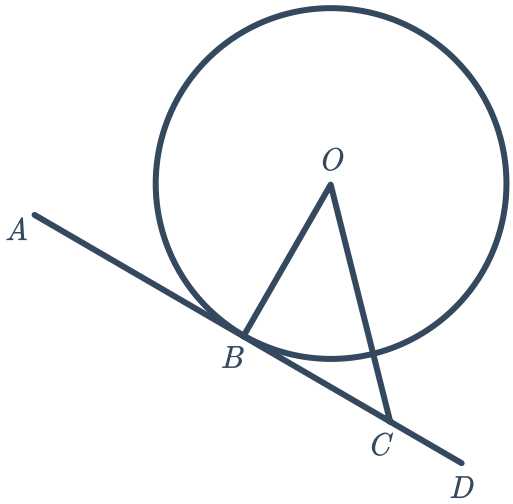
1 In the diagram, C is an arbitrary point on the line AD, and B is the point at which the tangent meets the circle.

a What can be said about the lengths of OB and OC?

b What point on AD is closest to the centre of the circle?

c In general, what can be said about the angle of a line joining some point to some other line by the shortest route?

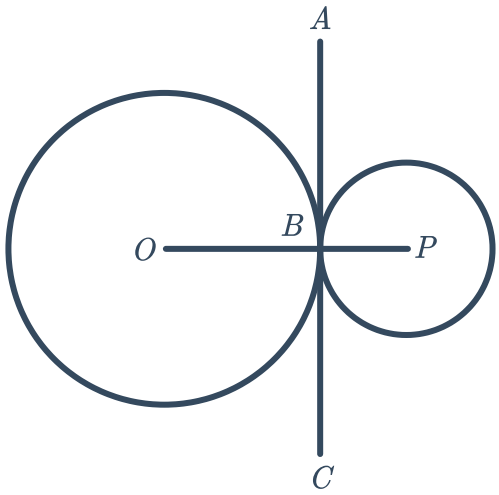
d Hence, what can be said about angle $\angle OBA$?



2 Consider the following figure:

a Show that $\angle OBP$ is a straight angle.

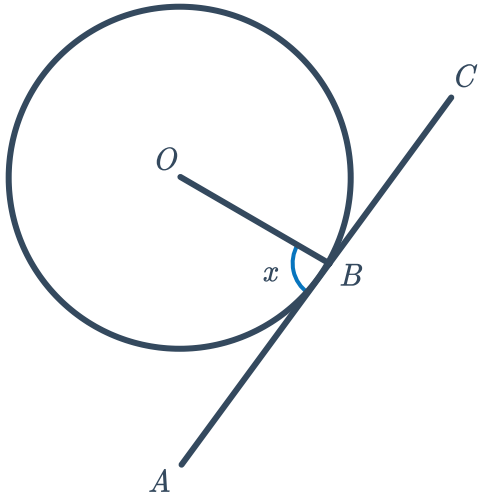
b Hence what can we say about points O, B and P?



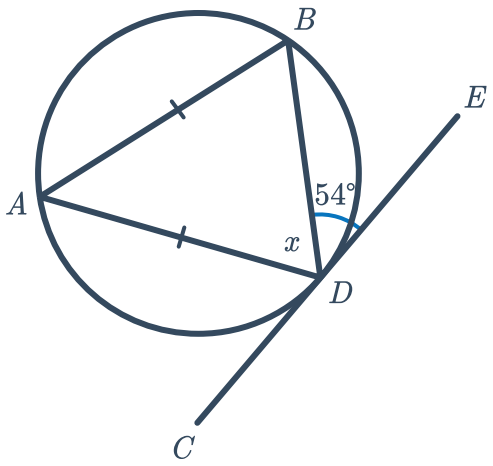
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3 Find the value of the pronumerals in the following diagrams. Give reasons for your answer.

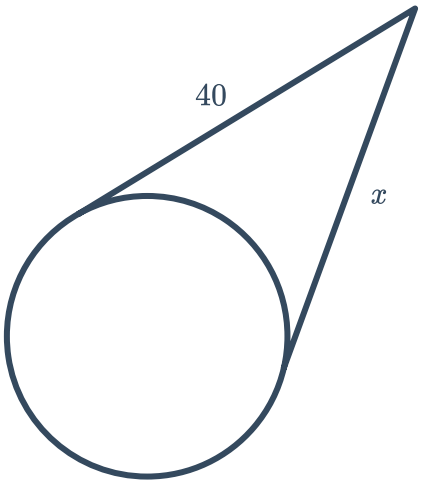
a



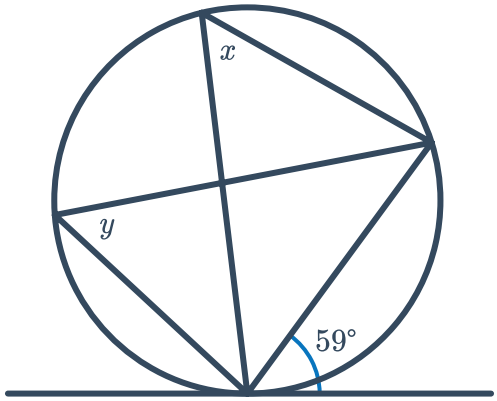
b



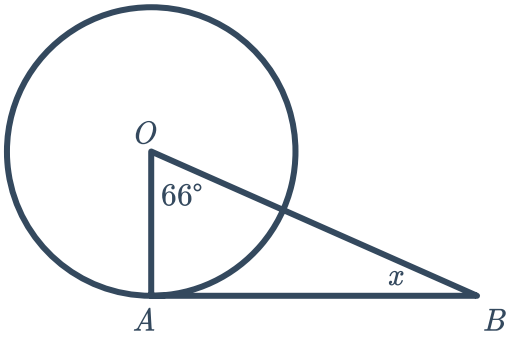
c



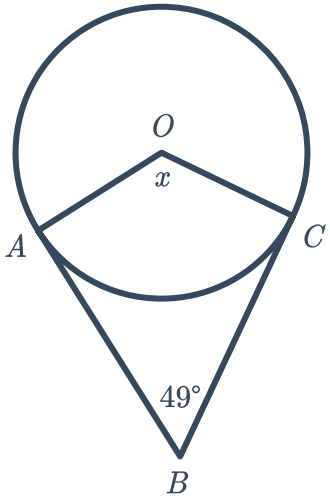
d



e



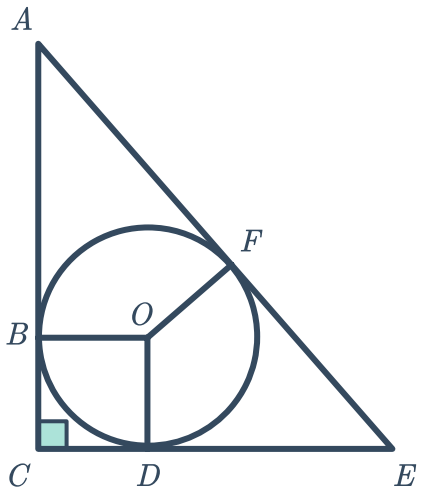
f



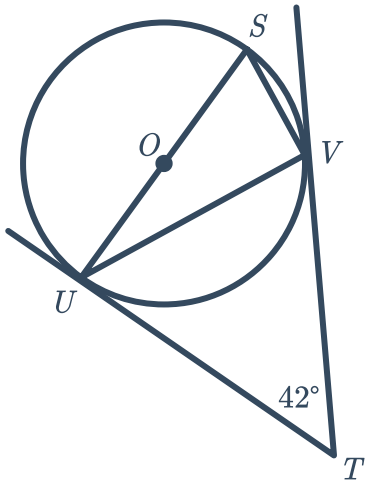
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4 The circle in the following diagram has a radius of 8 and AB has a length of 9 units.

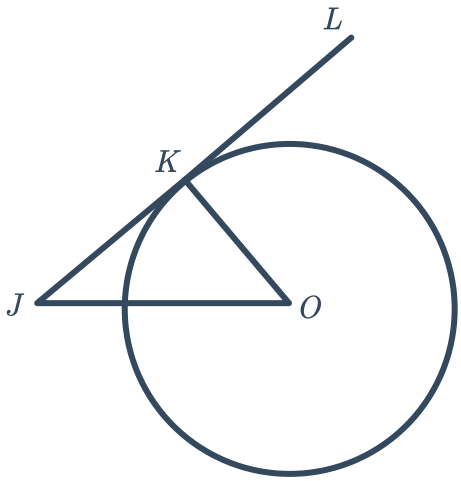
Calculate the length of FE.



5 In the diagram, TU and TV are tangents to the circle with centre O. Find the size of $\angle SUV$.



6 In the following diagram, JL is a tangent to the circle with centre O. $JO = 29$ and $JK = 20$. Find the length of the radius OK.

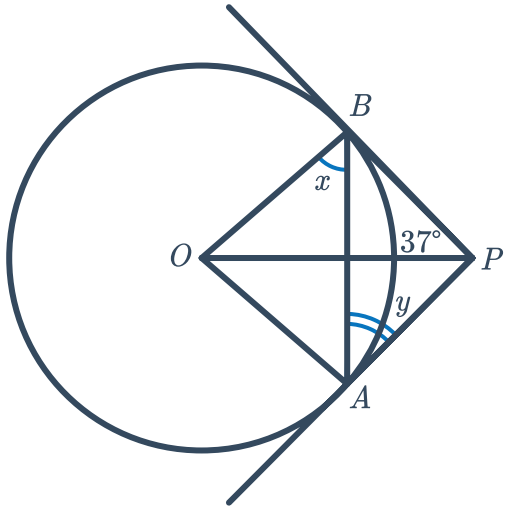


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7 In the following diagram, PA and PB are two tangents issued from point P to the circle of center O, such that $\angle PBO = 37$.

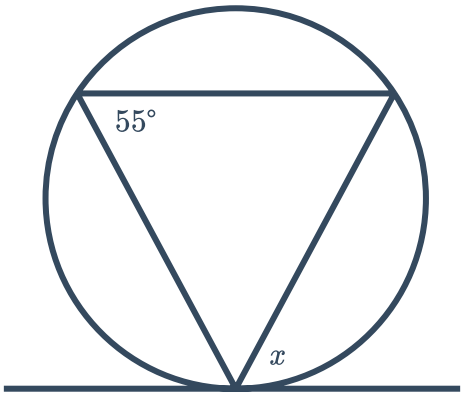
a Find the value of y.

b Find the value of x.

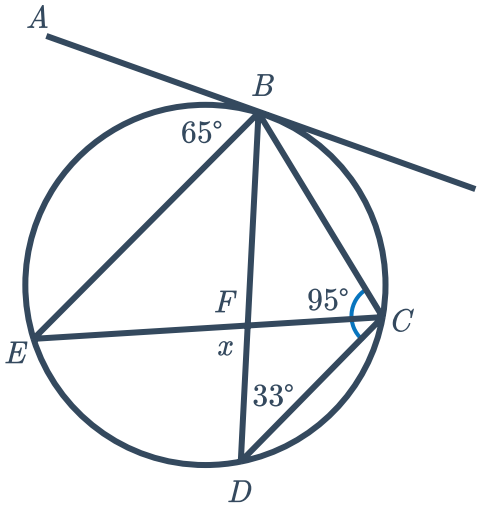


8 Find the value of the pronumerals in the following diagrams. Give reasons for your answer.

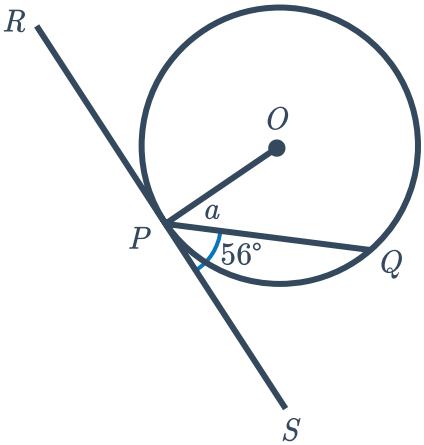
a



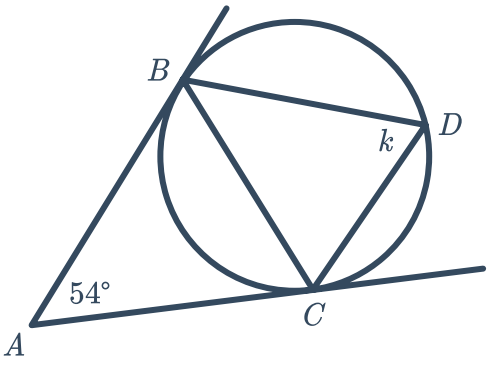
b



c

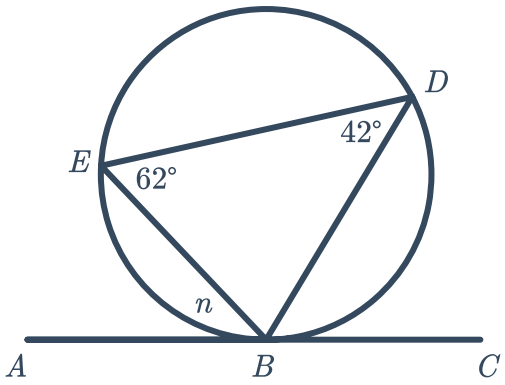


d

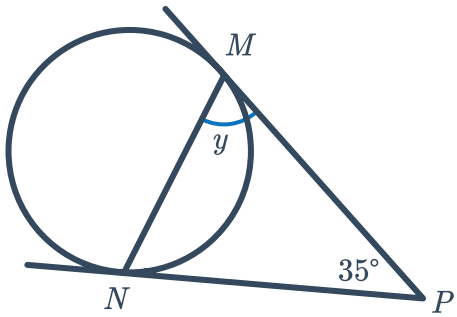


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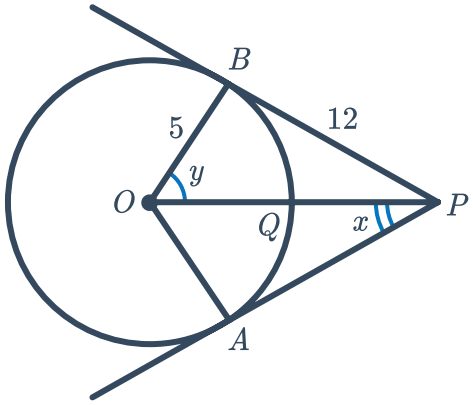
e



f



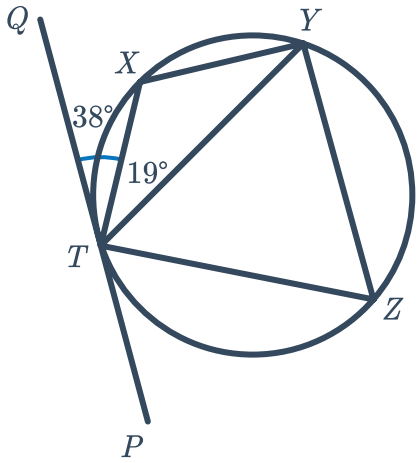
9 In the following diagram, PA and PB are two tangents issued from point P to the circle of center O, such that $PB = 5$.



a Find the value of x.

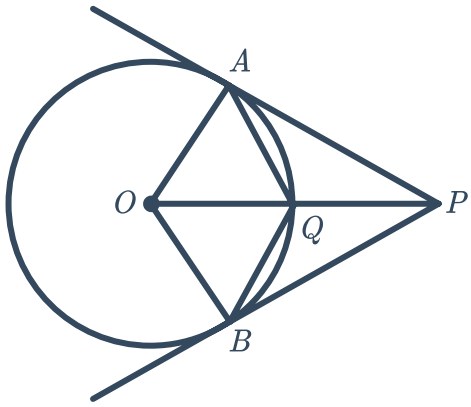
b Find the value of y.

10 In the diagram, PT is a tangent to the circle. Find the size of $\angle YZT$.

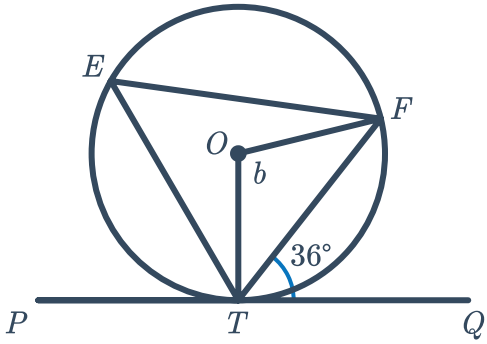


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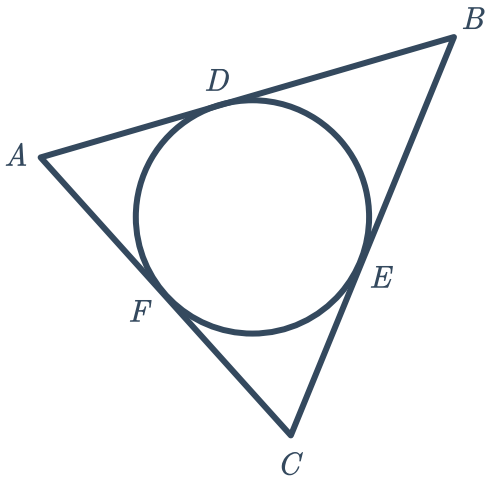
11 In the figure, O is the center of the circle, PA and PB are tangents. Find the size of $\angle APB$ if $\angle OAQ = 56^\circ$.



12 In the diagram, PT is a tangent to the circle with centre O. Determine the value of b.

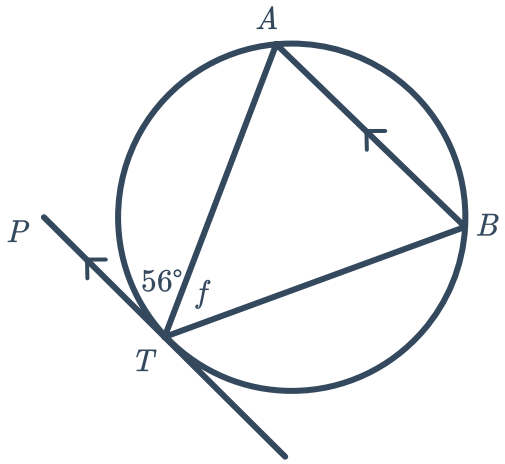


13 In the following diagram, AB, BC and AC are tangents to the circle. $BD = 7$ cm, $AF = 6$ cm, and $BC = 11$ cm. Find p, the perimeter of $\triangle ABC$.



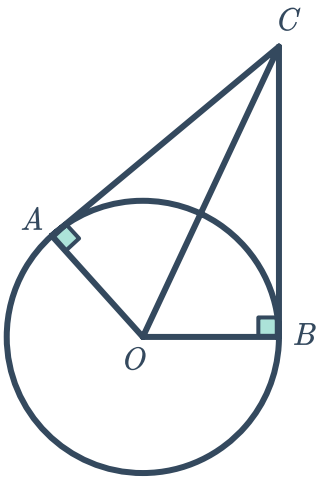
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14 In the diagram, PT is a tangent to the circle. Determine the value of f.

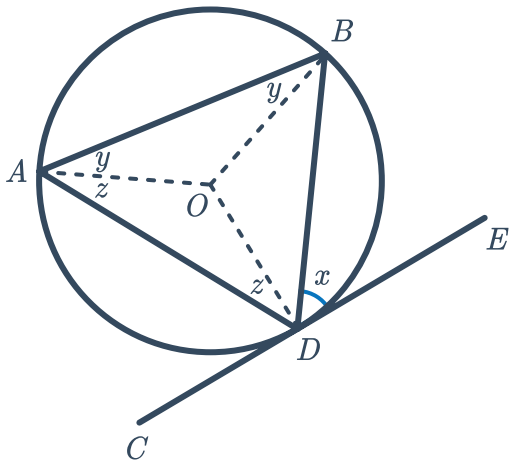


Proofs

15 Prove that $\triangle OAC$ and $\triangle OBC$ are congruent. Then, show that $AC = BC$.

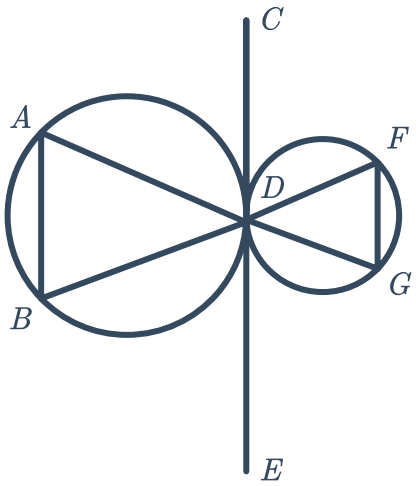


16 For the following diagram, prove that $y + z = x$.



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17 For the following diagram, prove that $AB \parallel FG$.



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